Multimodal Cross-Scene Hyperspectral Image Classification Guided by Dual-Agent Priori Knowledge

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***Abstract*—Semantic information is increasingly used to improve the performance of cross-scene hyperspectral image (HSI) classification. However, existing methods have difficulty fully exploiting the prior knowledge of large language models (LLMs) in textual information and usually rely on static template prompts. Moreover, they remain insufficient in modeling the complementary information between images and text. To address these issues, this article proposes a two-agent prior-guided multimodal network (TPMNet) for cross-scene HSI classification. First, a cross-domain semantic modeling mechanism guided by dual-agent priors is introduced. In this mechanism, source-domain and target-domain LLMs generate their respective prior texts. These priors are updated using similarity relationships with source-domain and target-domain image features, together with a memory-guided mechanism. As a result, dynamic textual representations with cross-domain perception capability are obtained. Second, a bidirectional multimodal cross-domain feature alignment and enhancement (BMAE) strategy is designed. Bidirectional cross-modal attention interactions are introduced between images and text to capture complementary discriminative representations across modalities. Meanwhile, a gating mechanism is adopted to select features that are highly relevant to the task, thereby improving the effectiveness of multimodal cross-domain features. Finally, a multiview consistency constraint learning (MCCL) method is proposed. Multilevel augmented views are constructed for target-domain samples. Multiple classifiers are designed to enforce joint consistency across different views, which enhances the classification capability of the model on target-domain data. Experimental results show that the proposed method achieves better performance than seven advanced algorithms on multiple cross-scene hyperspectral classification tasks.**

***Index Terms*—Domain adaptation, hyperspectral image (HSI), multimodal fusion, semantic prior learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain rich 3-D information [1]. In the spatial dimension, they can reflect the positional relationships among pixels. In the spectral dimension, hyperspectral data usually consist of tens to hundreds of spectral bands, which can characterize the physical properties of objects [2]. Benefiting from the complementary spatial and spectral information, hyperspectral images have been widely studied in fields such as military security [3], environmental monitoring [4], and agricultural inspection [5]. In traditional hyperspectral image classification tasks, the training set and the test set are usually assumed to be drawn from the same data distribution. With carefully designed network architectures and optimization strategies, deep learning-based methods have achieved remarkable performance in hyperspectral image classification [6]. However, in practical applications, the test data and the available training data often come from different distributions [7]. Therefore, performing hyperspectral image classification under cross-scene conditions has become a new and challenging problem [8].

Domain adaptation is an effective approach for addressing cross-scene classification tasks [2]. Its core idea is to jointly learn from labeled source-domain data and unlabeled target-domain data during training, and to perform classification on target-domain samples during testing. Traditional domain adaptation methods reduce the discrepancy between the source and target domains mainly through feature alignment. For example, Ganin and Lempitsky [9] proposed an adversarial feature alignment strategy by introducing a gradient reversal layer, while Zhang et al. [10] further aligned both the marginal and conditional distributions to achieve finer-grained domain alignment. Recently, the rapid development of large language models (LLMs) has provided new possibilities for cross-domain semantic modeling [11]. As a result, researchers have begun to incorporate textual modality information into domain adaptation methods to further enhance the cross-domain generalization capability of models.

After introducing the text modality into cross-scene tasks, a key challenge lies in how to utilize textual information to guide the model in reducing the domain gap. Transforming semantic knowledge from the text modality into discriminative representations that can effectively guide model generalization is crucial for achieving semantic-driven domain adaptation. In recent years, significant progress has been made in leveraging textual modality information to improve performance on cross-scene tasks and to enhance model generalization ability. Huang et al. [12] employed label semantics as textual

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information in order to effectively design a vision–language model for zero-shot generalization in HSI classification. Wang et al. [13] proposed aligning 3-D image features of HSIs with multigranularity textual features in order to effectively learn robust domain-invariant representations, achieving strong performance on target-domain data. Zhang et al. [14] constructed a semantic space consisting of coarse-grained and fine-grained textual descriptions and realized domain transfer through vision–language alignment. Wang et al. [15] designed global-level textual representations and patch-level textual representations, and performed cross-modal domain adaptation via a contrastive learning strategy. Wang et al. [16] fused semantic templates with image features to derive semantic-fused representations and subsequently achieved cross-scene classification by computing their similarity with image representations.

Another challenge introduced by incorporating the text modality lies in how to effectively fuse and align information from text and image modalities to construct more discriminative multimodal joint representations. Akewar and Chandak [17] investigated the role of language models and semantic information in advancing the development of general artificial intelligence in the remote sensing field. Since raw textual and visual information are distributed in different feature spaces, direct semantic alignment between the two modalities remains highly challenging [18]. To enable effective cross-modal collaboration, a variety of multimodal fusion methods have been proposed. Yang et al. [19] first used textual information to guide the fusion of remote sensing images, and then aligned text with visual feature representations to achieve joint classification of remote sensing images. Yu et al. [20] proposed a cross-modal feature matching network learning the correlated features between text and remote sensing images. Qin et al. [21] designed instance-level and manifold-level embedding spaces to model vision and language and to explore the distributional relationships across modalities. Cao et al. [22] extracted fused features across spatial–spectral dimensions and language dimensions using contrastive learning, achieving excellent performance on downstream tasks.

After obtaining effective cross-domain generalizable representations, consistency constraints can improve model performance by enhancing the model's ability to represent complex data distributions. Recently, introducing consistency constraints to improve the generalization capability of models on target-domain samples has attracted increasing attention in cross-scene scenarios. Hu et al. [23] effectively improved HSI classification performance by combining consistency regularization with a pseudolabel self-training strategy, without introducing complex mechanisms. Feng et al. [24] proposed an entropy-regularized dual-classifier constraint to enhance the consistency of feature distributions of target-domain data across different classifiers, achieving excellent performance on cross-domain datasets. Liu et al. [25] constructed multiview features for remote sensing images and designed a network to fuse multiview information, further improving the effectiveness of feature representations. Litrico et al. [26] enhanced the stability of target-domain predictions in unsupervised domain adaptation by introducing a reweighting strategy within the classifier loss function. Wang et al. [27] designed a dual-classifier framework that learns consistent target-domain data distributions through collaborative learning between multiple classifiers, thereby improving the stability of the learned target-domain representations.

Recently, Zhang et al. [28] proposed an anchor-guided discriminative subspace alignment and clustering framework for cross-scene hyperspectral imagery, extending cross-scene learning to the fully unsupervised setting through discriminative subspace alignment. Wang et al. [29] developed a multiscale spectral–spatial anchor graph fusion framework to effectively exploit complementary spectral–spatial information for multimodal remote sensing representation learning. The above studies have achieved significant progress in domain-adaptive cross-scene classification tasks by incorporating textual modalities, multimodal alignment strategies, and consistency constraints. In particular, existing works have demonstrated the effectiveness of the text modality in image-related tasks and have obtained meaningful key representations within multimodal alignment spaces. However, several limitations remain. First, designing textual modality information as static text prevents it from dynamically and effectively guiding cross-domain model generalization. Second, the learning of multimodal joint representations mainly focuses on one-directional modal alignment, lacking sufficiently rich representations enabled by bidirectional interactions between modalities. In addition, when further optimizing and enhancing the target-domain data distribution, existing methods lack joint modeling of semantic consistency between multiview features and multiclassifier predictions.

To address the above issues, this article proposes a dual-agent prior-guided multimodal network. To overcome the limitation that existing studies rely on static text templates and thus cannot dynamically and effectively guide cross-domain generalization, a dual-agent prior-guided cross-domain semantic modeling mechanism is introduced. In this mechanism, LLMs are employed to generate semantic prior texts for the source domain and the target domain, respectively. Since these initial texts still suffer from static representations and limited cross-domain generalization ability, the textual semantics are further dynamically updated by jointly considering the similarity weights between image features and textual semantics, as well as the historical weights accumulated in a memory bank. Subsequently, to address the lack of bidirectional fusion information in multimodal joint representations, a bidirectional multihead attention strategy is constructed. In this strategy, semantics are used as queries to compute responses from image features, while image features are used as queries to compute responses from semantic representations. A gating mechanism is then designed during the fusion to emphasize features that are highly relevant to the task. Furthermore, to alleviate the lack of collaborative optimization between multiview features and multiple classifiers under target-domain distribution consistency constraints, a multiview consistency constraint learning (MCCL) method is proposed. The entire network is trained using a domain-adaptive progressive strategy [30]. First, supervised learning is performed on labeled source-domain samples to establish

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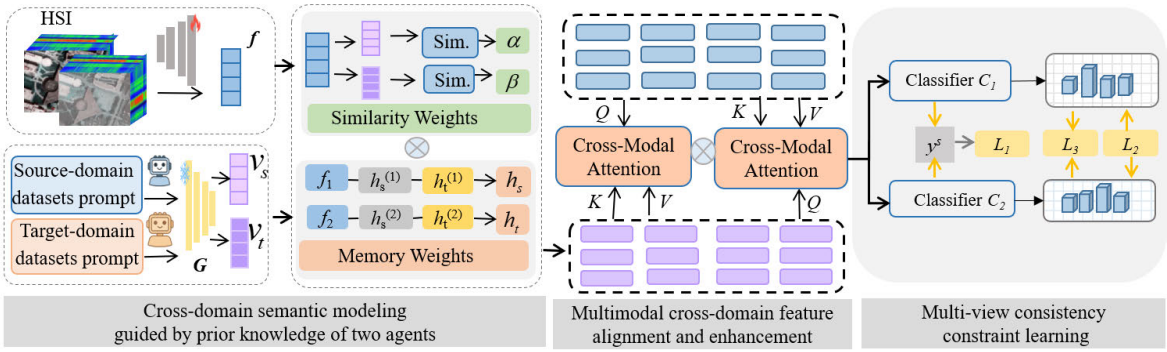


Fig. 1. Overall architecture of the proposed TPMNet.

basic hyperspectral discriminative capability. Then, the domain boundary is learned by maximizing the discrepancy between classifiers. Finally, multiview consistency constraints are introduced to optimize the unlabeled target-domain distribution more effectively, enabling accurate classification of target-domain data.

The main contributions of this article are as follows.

1) A dual-agent prior-guided cross-domain semantic modeling mechanism is proposed. This mechanism enables a transition of semantic representations from static templates to learnable priors, yielding dynamically evolving textual representations with cross-domain perception capability.
2) A multimodal cross-domain feature alignment strategy is developed. By employing bidirectional attention mechanisms and a gating design between modalities, this strategy aligns and enhances cross-modal representations, resulting in domain-invariant multimodal cross-domain joint features.
3) An MCCL method is proposed. The method constructs both strongly and weakly augmented views for target-domain samples, and enforces consistency among multiple classifiers on the prediction distributions of the same view, thereby improving the model's classification performance on the unlabeled target domain.

II. Methodology

A. Overall Framework

To address the limitations in multimodal cross-scene HSI classification, where the use of textual modality is restricted to static templates and cross-modal feature alignment lacks bidirectional learning, this article proposes a dual-agent prior-guided multimodal network. A dual-agent prior-guided cross-domain semantic modeling mechanism is designed, and a multimodal cross-domain feature alignment strategy is constructed to resolve the above issues. The overall framework of the proposed two-agent prior-guided multimodal network (TPMNet) is illustrated in Fig. 1.

As shown in Fig. 1, the proposed dual-agent prior-guided multimodal network consists of cross-domain semantic modeling guided by a dual-agent prior (SMAP) mechanism, which obtains prior textual information describing the characteristics of source-domain and target-domain data from the corresponding LLMs. These priors are then combined with similarity weights between the textual representations and image features, as well as memory weights, to produce dynamic semantics that can guide image features toward cross-domain generalization. Subsequently, the bidirectional multimodal cross-domain feature alignment and enhancement (BMAE) is constructed. Through bidirectional attention mechanisms between modalities, mutual alignment between image and text representations is achieved. Meanwhile, a gating mechanism is incorporated to emphasize features that are highly relevant to the task, resulting in fully aligned and selectively enhanced multimodal cross-domain joint representations. Finally, the MCCL is employed to enforce that multiple classifiers produce consistent prediction distributions on the same augmented view of target-domain samples. This enables the model to continuously accumulate stable self-guided learning on the unlabeled target domain, thereby enhancing its classification capability with respect to the target domain.

B. Cross-Domain Semantic Modeling Guided by Dual-Agent Prior

Existing studies still primarily rely on fixed static text templates when utilizing textual information. To overcome the limitation that static templates struggle to capture domain-invariant textual representations, this section designs a dual-agent prior-guided cross-domain semantic modeling mechanism. The mechanism consists of two agents corresponding to the source domain and the target domain, respectively. By associating textual information with image features through similarity weights and incorporating memory weights, cross-domain invariant textual representations are obtained.

In particular, the source domain agent and target domain agent are invoked to separately generate initial prior texts T_s and T_t for source-domain and target-domain image samples. The texts T_s and T_t are then fed into a text encoder G to obtain the initial source-domain semantic representation and the initial target-domain semantic representation, denoted as v_s and v_t, respectively. Subsequently, similarity weights and memory weights are computed for v_s and v_t to endow them with cross-domain consistent characteristics.

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First, the real-time similarity weights $[\alpha,\beta]$ between v_s, v_t and the source- and target-domain images are computed. These weights enable v_s and v_t to be dynamically adjusted through their interactions with source- and target-domain images, thereby facilitating the learning of text representations with cross-domain consistency, as formulated below

$$\alpha = \frac{\exp(v_s \cdot f)}{\exp(v_s \cdot f) + \exp(v_t \cdot f)} \tag{1}$$

$$\beta = \frac{\exp(v_t \cdot f)}{\exp(v_s \cdot f) + \exp(v_t \cdot f)} \tag{2}$$

where f denotes the feature of the current input image, extracted by the image encoder E. For each image sample, the sum of the weights α and β equals 1, which are used to measure the correlation between the source domain and target domain text embeddings and the image feature, respectively.

The proposed memory weight computation consists of two stages: memory addressing and memory aggregation. During training, two learnable memory banks are maintained, where the feature memory bank $B_f \in R^{M\times d}$ stores M historical feature representations and the text-domain preference memory bank $B_w \in R^{M\times 2}$ stores the corresponding historical text-domain preference. Both banks are jointly optimized through backpropagation.

The memory addressing stage retrieves historical feature representations that are relevant to the current image sample in the feature memory bank. In particular, the image feature f is projected through a projection matrix W_p to match the feature memory bank B_f. To eliminate the influence of feature magnitude, both f and B_f are normalized using the L_2 norm. The calculation is as follows:

$$\widetilde{f} = \frac{W_p f}{\|W_p f\|_2}, \quad m_i = \frac{B_f^{(i)}}{\|B_f^{(i)}\|_2} \tag{3}$$

where m_i denotes the ith normalized historical feature entry stored in B_f. The addressing weight is then computed according to the image feature and historical feature entry, as follows:

$$a_i = \frac{\exp\left(\widetilde{f} \cdot m_i^\top\right)}{\sum_{j=1}^{M} \exp\left(\widetilde{f} \cdot m_j^\top\right)}. \tag{4}$$

Next, the memory aggregation stage aggregates the corresponding historical text-domain pReferences stored in B_w using addressing weights as follows:

$$[h_s, h_t] = Softmax\left(\sum_{i=1}^{M} a_i B_w^{(i)}\right). \tag{5}$$

Therefore, the calculated memory weights $[h_s, h_t]$ capture historical text-domain preferences associated with the historical feature representations most relevant to the current image, thereby providing a global empirical prior for refining the text prior of the current image.

The final cross-domain dependence weights are jointly determined by the similarity weights and the memory weights. The former guides the textual representations toward domain consistency based on the similarity between textual representations from both domains and image features. The latter captures the model's long-term dependence preference for the two domain-specific texts. The final cross-domain dependence weights are computed accordingly as follows:

$$\omega_s = \frac{\alpha + h_s}{2} \tag{6}$$

$$\omega_t = \frac{\beta + h_t}{2} \tag{7}$$

where ω_s and ω_t represent the final cross-domain dependence weights of the source-domain and target-domain text representations, respectively, with their sum equal to 1. These weights are applied to the initial semantic representations of the source and target domains to further obtain the final semantic representation v as follows:

$$v = \omega_s v_s + \omega_t v_t. \tag{8}$$

In this section, two types of weights are designed while incorporating prior knowledge from LLMs, enabling the acquisition of textual representations with domain consistency under cross-scene settings and overcoming the limitation that static text templates struggle to support dynamic domain adaptation. Similarity weights enable the dynamic adjustment of text priors based on the immediate visual-semantic relevance of the current input sample. However, relying solely on instance-level similarity renders the alignment process susceptible to local error. To mitigate this limitation, memory weights introduce a global empirical constraint by capturing long-term text-domain preferences accumulated by the model during training. This fusion mechanism combines sample-specific responsiveness with historical experience to optimize the text prior, thereby achieving robust and domain-consistent text representations that help reduce the cross-domain gap between images.

Notably, the proposed dual-agent mechanism extends beyond the offline generation of textual priors by LLMs. Its dynamic nature is reflected in the adaptive refinement of text representations throughout the training process. During the HSI-to-text priors generation process, we construct prompts for each class within each dataset to instruct offline LLMs to describe the key spectral and visual features of that class for hyperspectral scenes of the source domain and target domain. In practice, both LLMs follow the same generation paradigm but are conditioned on different domain-specific prompts, resulting in two sets of domain-aware textual priors. During training, these textual priors are further refined by the proposed similarity-weighted and memory-guided strategy and are respectively aligned with source-domain and target-domain image features through bidirectional multimodal interaction. Consequently, their generated priors contribute differently to the final representation by providing domain-specific semantic guidance during cross-domain representation learning.

C. Bidirectional Multimodal Cross-Domain Feature Alignment and Enhancement

Existing studies on multimodal feature alignment strategies lack bidirectional attention interactions between modalities. To learn more discriminative multimodal joint representations, this section designs a multimodal cross-domain feature alignment strategy, including a bidirectional intermodal attention

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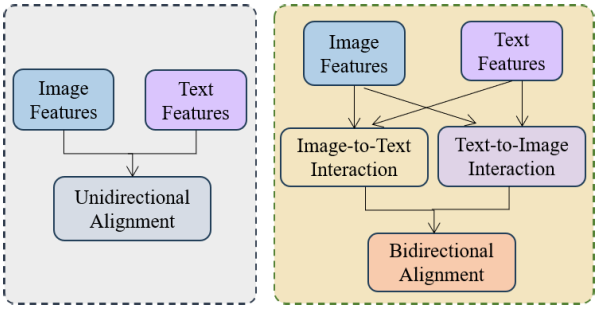


Fig. 2. Comparison diagram of the two alignment methods.

mechanism for alignment and a gating-based feature enhancement module. These components are used to obtain sufficiently aligned multimodal cross-domain joint representations and to emphasize features that are highly relevant to the task.

1) Bidirectional Alignment-Based Intermodal Attention Mechanism: Compared to unidirectional alignment, the proposed bidirectional multimodal interaction mechanism enables visual and textual representations to mutually enhance each other in a complementary manner. The comparison diagram of the two alignment methods is shown below.

As can be seen from Fig. 2, image-to-text interaction allows image features to selectively focus on the most relevant textual information, thereby mitigating interference from irrelevant semantic cues. Meanwhile, text-to-image interaction refines image features under the guidance of domain-consistent semantics. Through this bidirectional interaction, visual and textual features are progressively aligned within a shared semantic space, resulting in multimodal representations that are more discriminative and exhibit stronger domain consistency than those produced by unidirectional alignment. In particular, the image features f are used as the query, and the text features v serve as the key and value to compute self-attention, resulting in a text-centered multimodal joint representation h_{t-i}, detailed as below:

$$h_{t-i} = \frac{Soft\max(v \cdot f)}{\sqrt{d_f}} f \tag{9}$$

where d_f denotes the dimensionality of the image feature f and is used as a scaling factor to enhance training stability. Next, the hyperspectral textual feature v is used as the query, while the image feature serves as the key and value to compute self-attention, yielding an image-centered multimodal joint representation h_{i-t}, as follows:

$$h_{i-t} = \frac{Soft\max(f \cdot v)}{\sqrt{d_v}} v \tag{10}$$

where d_v denotes the dimensionality of the textual feature v and is likewise used as a scaling factor to enhance training stability. Subsequently, the text-centered multimodal representation and the image-centered multimodal representation are fused to obtain the bidirectionally aligned multimodal joint representation, as follows:

$$h_{fuse} = \beta h_{t-i} + (1-\beta) h_{i-t} \tag{11}$$

where $\beta \in [0, 1]$ serves as an adaptively optimized coefficient to control the fusion weights of the two representations.

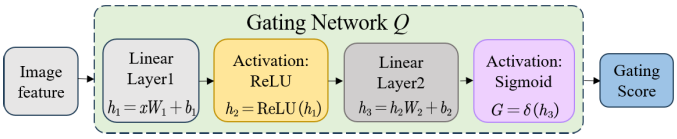


Fig. 3. Structural diagram of the gating network.

2) Feature Enhancement Based on a Gating Mechanism: After obtaining sufficiently aligned multimodal cross-domain features, a gating mechanism is designed in this section to further enhance the key joint feature representations under complex cross-domain shift scenarios.

Although the cross-modal representation incorporates complementary semantic information from text priors, the original visual representation still preserves valuable discriminative cues. Since the importance of these two representations may vary across samples, a lightweight gating network Q is introduced to adaptively balance their contributions and select task-relevant information. Its structure is shown in Fig. 3.

As illustrated in Fig. 3, Q takes the original image feature as input and consists of two fully connected layers with a ReLU activation, followed by a sigmoid function that predicts a gating scalar $g \in (0, 1)$. The enhanced multimodal features fused representation is computed as follows:

$$\hat{h} = g \cdot h_{fuse} + (1-g) \cdot f. \tag{12}$$

In this section, bidirectional intermodal attention learning is designed to obtain multimodal cross-domain joint representations, and a gating mechanism is constructed to enhance their salient information during the representation learning process. This design overcomes the limitation of single-modality alignment strategies that suffer from insufficient intermodal interactions. The multimodal cross-domain joint representations constructed for the source domain and the target domain in this section are then used as inputs to the classifiers to obtain prediction distributions, driving the model to perform robust classification under cross-domain conditions.

D. Multiview Consistency Constraint Learning

Existing studies adopt consistency constraint strategies to further drive models to learn the distribution of target-domain data, thereby improving classification performance on the target domain. However, current methods lack joint constraints on data distributions across multiple views and multiple classifiers. To address this issue, this section designs an MCCL scheme with multiple classifiers to enhance the model's classification performance on target-domain data.

Since target-domain data lack supervision from ground-truth labels during training, it is difficult for the model to directly and accurately capture their distribution. Therefore, this section constructs two augmentation operators to enable comprehensive learning of the target-domain data distribution from multiple views. In particular, the strong augmentation operator A_s consists of random horizontal flipping, random vertical flipping, and random noise, while the weak augmentation operator A_w consists of random horizontal flipping and random noise. The target-domain data processed by the strong

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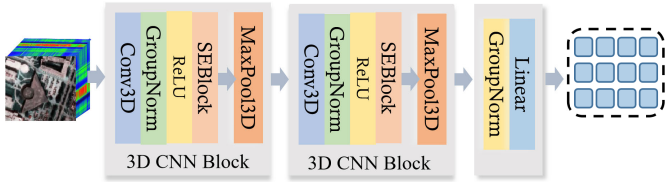


Fig. 4. Image encoder structure.

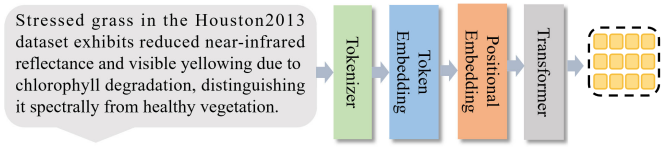


Fig. 5. Text encoder structure.

augmentation operator A_s and the weak augmentation operator A_w are denoted as x_s^t and x_w^t, respectively.

In this section, two classifiers, denoted as C_1 and C_2, are designed and constrained to produce similar predictive distributions for target-domain samples under the same view. This strategy encourages the model to progressively learn a more complete and accurate representation of the target-domain distribution. In particular, the predictive distributions of the two classifiers are first enforced to be consistent for target-domain samples under strong augmentation, which is formulated as follows:

$$L_s = KL\left(C_1\left(E\left(x_s^t\right), G\left(T_t\right)\right) \| C_2\left(E\left(x_s^t\right), G\left(T_t\right)\right)\right) \tag{13}$$

where $KL(\|)$ is used to measure the distance between two distributions and E and G denote the image encoder and the text encoder, respectively. Subsequently, the predictive distributions of the two classifiers are constrained to be consistent for target-domain samples under weak augmentation, which is computed as follows:

$$L_w = KL\left(C_1\left(E\left(x_w^t\right), G\left(T_t\right)\right) \middle\| C_2\left(E\left(x_w^t\right), G\left(T_t\right)\right)\right). \tag{14}$$

The final MCCL loss L_{cons} is obtained by jointly considering the constraints imposed by the two classifiers. Through the joint constraints across multiple classifiers and multiple views, this loss enhances the model's ability to correctly capture the target-domain data distribution, which is computed as follows:

$$L_{cons} = L_s + L_w. \tag{15}$$

In this section, a multiview consistency constraint loss is constructed. By designing multiple classifiers for the model and different augmentation operators for target-domain data, the model is encouraged to deeply learn the target-domain data distribution, thereby improving the effectiveness of existing methods that rely on a single constraint. Augmentation operators from different views generate variations at different spectral and noise levels, enabling the model to further explore the underlying patterns of target-domain data.

E. Training Optimization and Inference

TPMNet takes source-domain images and their labels, target-domain images, and the initial prior texts of both source-domain and target-domain data as inputs, and outputs the predicted labels for target-domain data. In particular, the image encoder is first randomly initialized. The source-domain and target-domain HSIs are then processed by the image encoder to obtain visual feature representations. The architecture of the image encoder E is illustrated in Fig. 4.

As shown in Fig. 4, the image encoder E is composed of multiple stacked 3-D CNN blocks. The 3-D convolutions are employed to simultaneously capture the global spectral responses and spatial structures of hyperspectral data. After convolution, group normalization is applied to enhance training stability, followed by ReLU to further improve nonlinear representation capability. Furthermore, an SEBlock is introduced to strengthen highly discriminative channel information. In addition, each 3-D CNN block incorporates a 3-D MaxPool layer to downsample the joint spectral–spatial representations, enabling the network to capture long-range feature information across layers. Finally, group normalization and a linear layer are used to project the image representations into a space aligned with the textual representations.

Next, the text encoder G transforms the prior texts generated by the dual agents into high-dimensional semantic embeddings that can be learned by the network. The workflow of the text encoder is illustrated in Fig. 5.

As illustrated in Fig. 5, the text encoder first converts the input text into a sequence of tokens using a tokenization layer. Subsequently, a token embedding layer maps the discrete token sequence into a continuous semantic space. To model the sequential relationships among semantic representations, a positional embedding layer is introduced, enabling the model to learn the contextual dependencies of semantic representations. The text sequence with positional information is then processed by Transformer layers to model and understand the high-level semantics of the textual descriptions.

After the prior descriptions of the source and target domains are encoded by the text encoder to obtain textual representations, similarity weights and memory weights are applied to endow them with cross-domain characteristics. Then, bidirectional modal alignment is performed between image features and text features, resulting in multimodal cross-domain joint representations. These representations from multiple perspectives are then fed into the classifier to produce the final category predictions.

The overall training process of TPMNet is divided into three stages [30]. In the first training stage, TPMNet performs supervised learning on the source domain using the labeled source-domain data, enabling the model to acquire classification capability for source-domain HSIs. The loss function of the first stage is defined as follows:

$$L_1\left(C_1, C_2, E\right) = C_1\left(E\left(x^s, y^s\right), G\left(T_s\right)\right) + C_2\left(E\left(x^s, y^s\right), G\left(T_s\right)\right) \tag{16}$$

where x^s and y^s denote the source-domain data and their corresponding labels, respectively. T_s and T_t represent the initial prior texts generated from the source-domain and target-domain image samples, respectively. C_1 and C_2 denote the two classifiers in TPMNet.

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TABLE I
NUMBER OF SAMPLES IN THE HOUSTON DATASET

No.	Class	Houston2013 (Source)	Houston2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

In the second training stage, the classification discrepancy between the two classifiers on the multimodal joint representations of the target-domain data is enlarged, enabling the model to gradually learn and adapt to the distribution of the target-domain data. The loss function of this stage is defined as

$$L_2(C_1, C_2) = -\left\| C_1\left(E\left(x^t\right), G(T_t)\right) - C_2\left(E\left(x^t\right), G(T_t)\right) \right\|. \tag{17}$$

In the third stage, building upon the second stage, an MCCL loss is introduced to enable the model to obtain discriminative self-supervised signals from multiview representations. In particular, by enforcing multiple classifiers to produce highly consistent predictive distributions for target-domain samples under the same view, the classification accuracy of the model is further enhanced.

$$L_3(E) = L_{cons} - L_2. \tag{18}$$

After the above three training stages, TPMNet acquires the capability to classify both source-domain and target-domain HSIs by optimizing multimodal cross-domain joint representations and leveraging MCCL. During the testing stage, TPMNet does not utilize textual data, and the classifiers directly perform prediction based on the outputs of the image encoder for the target-domain images.

III. EXPERIMENTAL ANALYSIS

A. Datasets

1) Houston Dataset: This dataset consists of two HSIs acquired over the Houston area in the United States in 2013 and 2018, corresponding to Houston2013 [31] and Houston2018 [32], respectively. To conduct cross-scene experiments, the overlapping region and the common spectral bands between the two datasets are used. The class distribution is summarized in Table I.

2) S–H Dataset: This dataset consists of HSIs collected from Shanghai and Hangzhou, respectively. The data were acquired by the EO-Hyperion sensor. In the experiments, only the shared spectral bands and classes between the two regions are retained. The class statistics are reported quantitatively in Table II.

TABLE II
NUMBER OF SAMPLES IN THE S–H DATASET

No.	Class	Hangzhou (Source)	Shanghai (Target)
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40307	83188
	Total	135700	368000

TABLE III
NUMBER OF SAMPLES IN THE YC–YC DATASET

No.	Class	ZY1-02D Yancheng (Source)	GF5 Yancheng (Target)
1	Architecture	451	360
2	River	214	217
3	Reed	244	132
4	Reed	3026	832
5	Fallow land	650	234
6	Sea	2076	2395
7	Offshore water	1558	1305
	Total	8219	5475

3) YC–YC Dataset: This dataset is composed of two hyperspectral datasets, namely ZY1-02-D Yancheng and GF5 Yancheng [33]. The data were acquired over the Yancheng wetlands by advanced hyperspectral imagers onboard China's Ziyuan-1-02-D and Gaofen-5 satellites. Seven land-cover classes and 147 shared spectral bands are selected for cross-scene studies. The class statistics are summarized in Table III.

B. Comparison Methods

To evaluate the proposed method, seven state-of-the-art cross-scene domain adaptation algorithms for HSI classification are selected for comparison. DSAN [34] reduces the distribution discrepancy between different domains by minimizing statistical distances. FDDAN [35] enhances the modeling of target-domain distribution details through a progressive training strategy that jointly exploits class-level and domain-level features. SCLUDA [36] introduces a domain similarity loss to alleviate domain shift in cross-scene HSIs. TSTnet [37] achieves distribution alignment across domains by aligning topological relationships modeled by graph structures. MLUDA [38] mitigates the domain gap from multiple perspectives, including image features and distribution characteristics of both source and target domains. CSGNet [39] leverages class-related semantic information to obtain domain-invariant representations. TASA-Net [16] designs a two-level semantic representation with coarse-grained and fine-grained semantics to facilitate cross-scene transfer. All experiments are conducted using an RTX 2080 Ti GPU and the PyTorch framework. A total of 190 source-domain samples together with all target-domain samples are used for training, and evaluation is performed on the entire target domain.

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TABLE IV
CLASSIFICATION ACCURACY OF EIGHT METHODS ON THE HOUSTON DATASET

Class	DSAN	FDDAN	TSTnet	SCLUDA	MLUDA	CSGNet	TASA-Net	TPMNet
1	88.60	97.07	86.13	82.45	68.46	52.93	31.40	71.35
2	62.89	26.15	69.52	61.58	78.04	41.73	59.41	83.30
3	67.76	63.43	63.80	59.34	58.51	43.65	57.14	60.52
4	90.00	95.00	98.64	45.45	92.73	79.09	68.18	100.00
5	54.39	73.70	79.58	86.53	81.90	53.00	84.07	80.95
6	49.05	64.35	76.62	81.62	79.78	93.48	79.73	80.70
7	55.07	40.00	54.26	45.82	53.22	9.66	33.42	58.52
OA	59.08±2.58	59.67±4.26	73.17±2.58	74.84±1.17	75.25±2.16	70.96±2.17	70.33±1.55	**77.01±1.37**
AA	66.82±3.19	65.67±4.88	75.50±1.46	66.11±4.27	73.23±4.91	53.36±6.53	59.05±6.35	**76.48±1.52**
Kappa	43.29±3.16	41.51±5.52	58.80±3.16	59.85±1.89	60.69±4.12	42.89±6.61	50.69±4.22	**63.42±1.63**

C. Evaluation Indicators

In this study, overall accuracy (OA), average accuracy (AA), and the Kappa coefficient are adopted as evaluation metrics to assess the performance of algorithms [40]. OA reflects the proportion of correctly classified samples over the entire test set. AA computes the average classification accuracy across all classes, thereby providing a more comprehensive evaluation of per-class performance. The Kappa coefficient accounts for the expected agreement under random classification, and thus can more reliably reflect the true classification capability of a model under imbalanced class distributions, effectively compensating for the limitations of OA. These three metrics are widely used in cross-scene HSI classification and provide a comprehensive evaluation of classification performance.

D. Experimental Setup

For DSAN, the weights of the supervised loss and the adaptation loss are set to 1 and 0.5, respectively. For FDDAN, the learning rates for the three datasets are set to 0.01, 0.001, and 0.01. For SCLUDA, the optimizer is stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 5×10^{-4}. For TSTnet, all three datasets are trained for 500 epochs. For MLUDA, the patch sizes for the three datasets are set to 7, 3, and 7, respectively. For both CSGNet and TASA-Net, the text encoders are initialized with the pretrained ViT-B-32.pt model. For the proposed TPMNet, the number of training epochs is set to 100 for the three datasets, and SGD is used as the optimizer. The initial learning rates for the three datasets are set to 0.01, 0.01, and 0.05, respectively. The learning rate scheduling strategy for the three datasets is defined as $\eta_s = \eta_0 \cdot (1 + \lambda \cdot s)^{-\gamma}$, in which the learning rate at the sth training step is denoted by η_s. η_0 represents the initial learning rate, λ is the scheduler control coefficient with a value of 2×10^{-4}, γ and is the decay factor with a value of 0.75. The batch size for three datasets is set to 32.

For the proposed TPMNet, the LLM Qwen-Plus is employed as an agent to generate prior textual descriptions of the visual and spectral features for both source and target domains, based on a unified prompt design. This model is solely used for producing prior information and does not participate in network training. The prompt takes the form: "Please describe the key spectral and visual features of {class_name} in the {source_data_name} from the {cross-scene_data_name}." The prior texts, which contain both visual and spectral descriptions, are subsequently fed into the pretrained text encoder ViT-B-32.pt to obtain initial semantic representations, supporting the cross-domain classification task.

E. Experimental Results and Analysis

1) Performance on the Houston Dataset: For the Houston dataset, Table IV presents the detailed classification performance of the seven comparison methods and the proposed TPMNet on the target domain.

As shown in Table IV, compared with the seven state-of-the-art methods, TPMNet achieves the best performance. In terms of OA, MLUDA exceeds 75%, whereas TPMNet further improves the performance to 77.01%, yielding a gain of 1.76% over the second-best method. For AA, TSTnet shows competitive performance with 75.50%, while TPMNet attains the highest accuracy of 76.48%, indicating the best class-averaged performance. MLUDA reduces interdomain discrepancies through multilevel cross-domain alignment and achieves the second-best Kappa value of 60.69%. TPMNet further improves this value to 63.42%. Fig. 6 illustrates the prediction maps of the comparison methods and the proposed TPMNet on the target domain of the Houston dataset.

As shown in Fig. 6, for the Houston target-domain data, the overall classification map produced by the proposed TPMNet shows the highest consistency with the ground truth, fully demonstrating the model's reliable cross-scene classification capability. In the regions highlighted by the white boxes, the residential buildings and road classes are interleaved, making classification particularly challenging. Most methods tend to misclassify residential buildings as the visually similar non-residential buildings class. However, both TSTnet and TPMNet are able to classify residential buildings more accurately. Moreover, for the elongated road class, TPMNet produces clearer and more precise predictions. This advantage can be attributed to TPMNet's bidirectional cross-modal attention mechanism and the adaptive gating network, which effectively leverage complementary information from both textual and visual modalities. By dynamically balancing the contributions of different modalities, the model learns

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TABLE V
CLASSIFICATION ACCURACY OF EIGHT METHODS ON THE S–H DATASET

Class	DSAN	FDDAN	TSTnet	SCLUDA	MLUDA	CSGNet	TASA-Net	TPMNet
1	94.80	92.60	96.61	94.74	95.10	93.24	93.08	93.40±0.07
2	68.34	85.52	86.00	82.30	78.94	89.18	85.22	89.48±0.05
3	99.95	98.62	83.15	99.68	96.40	89.30	98.50	100.00±0.00
OA	84.34±4.34	90.85±1.44	88.91±4.53	90.39±2.12	88.29±6.03	90.56±3.71	90.85±1.17	**93.17±0.07**
AA	87.70±3.33	92.25±1.10	88.59±7.51	92.24±1.60	90.15±5.92	90.57±6.28	92.27±0.94	**94.29±0.05**
Kappa	76.76±6.11	86.02±2.15	82.67±7.65	85.42±3.10	82.31±8.90	85.27±6.25	86.02±1.74	**89.51±0.10**

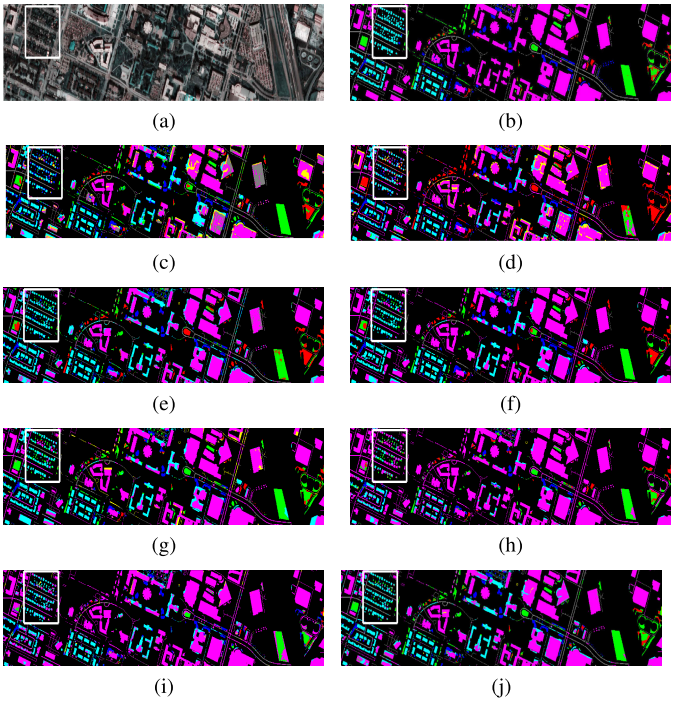


Fig. 6. Classification results of the Houston dataset. (a) False-color image. (b) Ground truth image. (c) DSAN. (d) FDDAN. (e) TSTnet. (f) SCLUDA. (g) MLUDA. (h) CSGNet. (i) TASA-Net. (j) TPMNet.

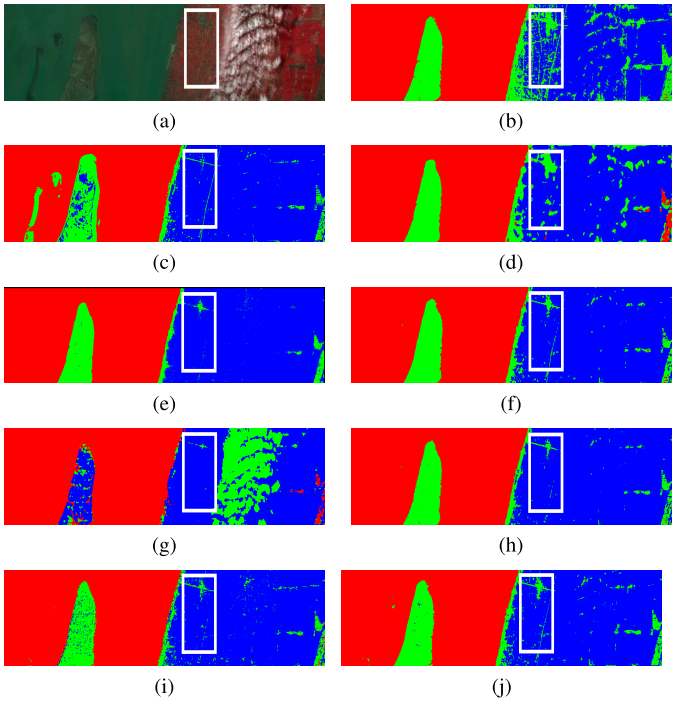


Fig. 7. Classification results of the S–H dataset. (a) False-color image. (b) Ground truth image. (c) DSAN. (d) FDDAN. (e) TSTnet. (f) SCLUDA. (g) MLUDA. (h) CSGNet. (i) TASA-Net. (j) TPMNet.

more robust and discriminative cross-domain hyperspectral representations. Consequently, TPMNet achieves stronger generalization ability and superior classification performance on the unlabeled target-domain data.

2) Performance on S–H Dataset: For the S–H dataset, Table V presents the detailed classification performance in a comprehensive manner of the seven comparison methods and the proposed TPMNet on the target domain. Bold the optimal values and underline the suboptimal values.

As shown in Table V, compared with the seven state-of-the-art methods, TPMNet achieves the best performance across all three metrics on the S–H dataset. For OA, TASA-Net achieves the second-best value of 90.85% under the guidance of coarse- and fine-grained textual information, whereas TPMNet, guided by the dual agents with a memory bank, reaches the highest accuracy of 93.17%. This substantial improvement demonstrates the effectiveness of the proposed dual-agent prior-guided semantic modeling and bidirectional multimodal fusion strategy. For AA, TASA-Net attains the second-best result of 92.27%, while TPMNet further improves the class-averaged performance to 94.29%. This indicates that TPMNet not only excels in overall classification but also achieves more balanced and robust results across different land cover categories. In terms of the Kappa coefficient, TASA-Net obtains the second-best value of 86.02%. By leveraging the multiview consistency constraint mechanism proposed in this article, TPMNet mitigates uncertainty in cross-scene scenarios and further achieves the highest Kappa of 89.51%. These consistent gains across OA, AA, and Kappa fully validate the superiority of TPMNet in addressing domain shift challenges. Fig. 7 illustrates the prediction maps on the target domain.

As shown in Fig. 7, the proposed TPMNet demonstrates strong cross-domain classification capability on the S–H target domain, with the highest consistency between its prediction map and the ground truth. Since the water class is densely distributed, all methods achieve good classification performance for this class. In the regions highlighted by the white boxes, the classification is more challenging due to a small

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TABLE VI
CLASSIFICATION ACCURACY OF EIGHT METHODS ON THE YC–YC DATASET

Class	DSAN	FDDAN	TSTnet	SCLUDA	MLUDA	CSGNet	TASA-Net	TPMNet
1	46.64	70.72	31.92	59.86	45.89	76.91	68.02	83.44
2	34.33	44.06	32.44	22.86	21.67	96.59	84.28	98.66
3	66.67	79.32	42.05	9.70	10.76	51.00	29.29	63.58
4	85.32	94.54	56.03	91.75	97.29	97.84	96.92	97.33
5	32.65	50.56	0.77	26.67	55.28	17.01	17.09	1.11
6	57.66	31.47	75.31	93.34	88.01	90.48	60.63	99.62
7	99.23	88.81	72.02	97.59	92.00	99.93	96.52	99.99
OA	69.27±10.60	59.77±10.54	63.05±19.58	84.25±8.48	80.24±7.41	89.06±12.40	73.47±14.18	**93.13±0.77**
AA	60.36±8.63	65.64±7.68	44.36±16.81	57.40±4.08	58.70±4.43	75.68±8.01	64.68±6.86	**77.68±1.80**
Kappa	61.86±11.76	52.38±11.58	49.73±25.43	78.81±9.72	75.69±9.25	86.09±13.59	67.04±16.00	**90.53±1.06**

number of land/building samples being interleaved with a large number of plant samples. Most methods fail to accurately identify land/building samples in these regions. On the contrary, FDDAN and TPMNet are able to recognize more land/building samples. Moreover, for the elongated land/building class, TPMNet provides more precise predictions. This advantage primarily benefits from TPMNet's unsupervised multiview consistency learning on target-domain data, which enables the model to better capture the distribution of target-domain samples and thereby enhances its classification capability.

3) Performance on the YC–YC Dataset: For the YC–YC dataset, Table VI presents the detailed classification performance of the seven comparison methods and the proposed TPMNet on the target domain.

As shown in Table VI, compared with the seven state-of-the-art methods, TPMNet achieves the best performance across all three metrics. CSGNet leverages class description texts to enhance the feature representation of each class and achieves an OA of 89.06%. TPMNet constructs cross-domain consistent dynamic text representations and reaches an OA of 93.13%. For AA, CSGNet attains the second-best value of 75.68%, while TPMNet achieves the highest accuracy of 77.68%. The fifth class is more challenging due to its fragmented distribution, resulting in lower classification accuracy across all methods. TPMNet achieves the best performance on classes 1, 2, 6, and 7. CSGNet also achieves the second-best Kappa of 86.09%, whereas TPMNet attains the highest Kappa of 90.53%. Fig. 8 illustrates the prediction maps of the comparison methods and the proposed TPMNet on the target domain of the YC–YC dataset.

As can be seen from Fig. 8, among the eight compared state-of-the-art cross-scene HSI classification methods, the classification results generated by TPMNet exhibit higher overall structural consistency with the corresponding ground truth maps, demonstrating stronger cross-domain classification capability. In the regions highlighted by white boxes, the classes architecture, river, and paddy show unclear boundaries and scattered spatial structures, posing significant challenges for classification. In these complex areas, FDDAN and TPMNet are able to distinguish the different classes more effectively. For the more regularly distributed sea class, FDDAN tends to misclassify it as river, whereas TPMNet provides more accurate predictions. These results indicate that TPMNet not only has strong discriminative ability in regular regions but also demonstrates greater robustness in the complex overlapping-class areas.

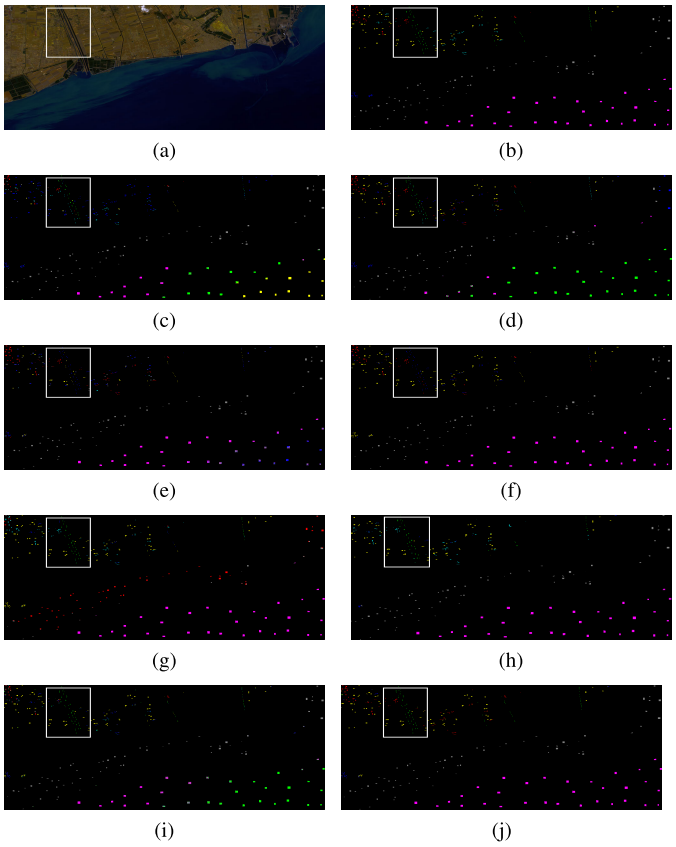


Fig. 8. Classification results of the YC–YC dataset. (a) False-color image. (b) Ground truth image. (c) DSAN. (d) FDDAN. (e) TSTnet. (f) SCLUDA. (g) MLUDA. (h) CSGNet. (i) TASA-Net. (j) TPMNet.

F. Ablation Study

This section presents ablation experiments to further assess the effectiveness of each module. In TPMNet, the SMAP is examined by its exclusion, which demonstrates the contribution of cross-domain consistent dynamic semantics. The role of the BMAE is evaluated by omitting it, highlighting the effectiveness of the intermodal attention fusion mechanism. Similarly, the effect of the MCCL is investigated by its removal, showing the importance of applying multiview

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TABLE VII
OA IN THE ABLATION STUDY ON THE THREE DATASETS

SMAP	BMAE	MCCL	Houston	S-H	YC-YC
	✓	✓	75.43±0.86	91.90±0.15	89.60±9.38
✓		✓	75.32±1.17	91.92±0.10	91.80±0.04
✓	✓		74.45±0.62	92.49±0.06	91.94±0.17
✓	✓	✓	**77.01±1.37**	**93.17±0.07**	**93.13±0.77**

TABLE VIII
EFFECTS OF SIMILARITY AND MEMORY WEIGHTS

w/o Sim.	w/o Mem.	Houston	S-H	YC-YC
✓		76.28 ± 2.07	91.86 ± 0.06	92.64 ± 1.04
	✓	76.87 ± 1.21	91.81 ± 0.17	92.66 ± 1.46
✓	✓	**77.01 ± 1.37**	**93.17 ± 0.07**	**93.13 ± 0.77**

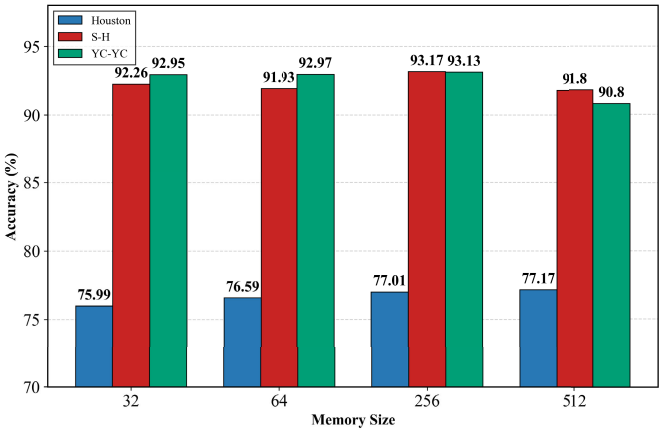


Fig. 9. Effect of memory bank size for different cross-scene tasks.

and multiclassifier constraints. The experimental results are summarized in Table VII.

The results in Table VII demonstrate the effectiveness of each proposed component. For the Houston dataset, removing the SMAP leads to a decrease of 1.58% in OA, omitting the BMAE results in a 1.69% drop, and removing the MCCL causes a 2.56% reduction. Similarly, for the S–H dataset, exclusion of SMAP reduces the accuracy by 1.27%, BMAE removal decreases it by 1.25%, and MCCL removal results in a 0.68% drop. For the YC–YC dataset, removing SMAP leads to a 3.53% decrease, BMAE removal causes a 1.33% reduction, and MCCL removal results in a 1.19% drop in accuracy. These ablation experiments confirm that each proposed module contributes positively to the performance of TPMNet, validating the effectiveness of the individual innovations as well as the overall network.

G. Analysis of the Effects of Similarity and Memory Weights on Cross-Domain Semantic Modeling

This section experimentally examines the roles of similarity and memory weights in dynamic text representation learning. Based on TPMNet, two variant models are constructed: w/o similarity weight and w/o memory weight. In the former, the similarity relationship between text embeddings and image features is no longer considered during text representation updates. In the latter, the memory-guided mechanism is removed. The experimental results are presented in Table VIII.

As observed in Table VIII, removing the similarity weight leads to a performance drop across all datasets, indicating that the similarity-guided semantic alignment mechanism is crucial for constructing dynamic text representations with cross-domain awareness. Meanwhile, removing the memory weight also results in decreased performance, demonstrating that the memory-guided mechanism plays an active role in maintaining cross-domain semantic consistency. The synergistic effect of both mechanisms enables the model to build dynamic and cross-domain-aware text representations, thereby enhancing the multimodal learning performance in cross-scene HSI classification tasks.

H. Sensitivity Analysis of Memory Bank Size M

To evaluate the impact of different memory sizes in the proposed memory mechanism, we conduct a sensitivity analysis on memory size. We set the memory size to 32, 64, 256, and 512, respectively, with experimental results shown in Fig. 9.

Fig. 9 demonstrates that TPMNet exhibits strong robustness to memory size. The memory bank maintains the historical contributions of source and target domain text representations to model optimization, with its size directly determining the degree of utilizing historical domain preference information. In particular, on the Houston dataset, classification accuracy steadily increased from 75.99% to 77.17% as the memory size scaled from 32 to 512. This indicates that in the cross-domain scenario of the Houston dataset, where the data collection years in the source and target domains are significantly separated, a larger memory capacity helps the model accumulate more long-term textual domain preference information, thereby gradually enhancing its cross-domain perception capabilities. When the memory size is 256, the model achieves optimal performance on both the S–H and YC–YC datasets. Conversely, excessively large memory sizes may introduce redundancy or noise, leading to a slight performance decline. For the S–H dataset, accuracy increased from 92.26% to 93.17% when the memory size is scaled from 32 to 256, then slightly decreased to 91.80% at size 512. Similarly, for the YC–YC task, accuracy peaked at 93.13% with a size of 256 and decreased to 90.80% at size 512.

Finally, we uniformly adopt a memory size of 256 for three datasets. This setting achieved optimal or near-optimal performance while maintaining moderate memory consumption and training efficiency, further validating the robustness and practicality of the memory-guided mechanism.

I. Analysis of the Computational Complexity

TPMNet mainly consists of SMAP, BMAE, and MCCL. The SMAP mainly involves similarity computation and memory bank retrieval. Let d denote the feature dimension, M the memory bank size, and B the batch size. The similarity weighting requires $O(Bd)$ operations, while the memory-guided refinement introduces $O(BMd)$ complexity. The BMAE

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TABLE IX
TRAINING TIME OF ALL THE ALGORITHMS AT ONE EPOCH (SECONDS)

Datasets	DSAN	FDDAN	TSTnet	SCLUDA
Houston	8.29	18.14	4.43	3.76
S-H	1.19	6.14	2.11	1.11
YC-YC	4.99	16.82	6.00	4.49
Datasets	MLUDA	CSGNet	TASA-Net	TPMNet
Houston	2.52	24.82	8.62	11.09
S-H	1.27	5.91	2.13	3.36
YC-YC	1.88	13.23	9.14	9.92

TABLE X
INFERENCE TIME OF ALL THE ALGORITHMS ON THE TEST SET (SECONDS)

Datasets	DSAN	FDDAN	TSTnet	SCLUDA
Houston	110.14	6.4	10.43	5.59
S-H	211.23	66.49	321	38.05
YC-YC	6.87	0.89	5.58	0.8
Datasets	MLUDA	CSGNet	TASA-Net	TPMNet
Houston	7.77	16.49	81	10.19
S-H	65.53	121.37	292.85	73.67
YC-YC	0.69	1.44	8.63	1.02

performs two standard cross-attention operations for bidirectional image-text interaction followed by a lightweight gating network. The additional complexity of this module is approximately $O(BLd^2)$, where L is the fixed text sequence length. The MCCL introduces strongly and weakly augmented target samples, incurring the complexity of $O(BC)$, where C is the number of classes.

To further evaluate the technical complexity of the TPMNet, we compare it with state-of-the-art methods in terms of per-epoch training time. The training times on the three cross-scene datasets are reported in seconds in Table IX.

As shown in Table IX, TPMNet exhibits moderate training times across all datasets, achieving a good balance between computational efficiency and classification performance. In particular, on the Houston dataset, TPMNet requires 11.09 s/epoch, which is higher than the lightweight methods MLUDA at 2.52 s and SCLUDA at 3.76 s, but significantly lower than CSGNet at 24.82 s and FDDAN at 18.14 s. Considering that TPMNet substantially outperforms these baselines in accuracy, the additional computational overhead is reasonable. On the S–H dataset, TPMNet consumes 3.36 s/epoch, placing it in the midrange among the compared methods. It is faster than CSGNet at 5.91 s and FDDAN at 6.14 s while remaining competitive with the faster SCLUDA at 1.11 s and MLUDA at 1.27 s. On the YC–YC dataset, TPMNet requires 9.92 s/epoch, comparable to TASA-Net at 9.14 s, and notably lower than FDDAN at 16.82 s and CSGNet at 13.23 s.

Overall, even after incorporating the dual-agent prior guidance and dynamic memory mechanisms, TPMNet maintains reasonable training overhead. These components enable superior cross-domain classification performance without excessively increasing the computational burden. Therefore, TPMNet demonstrates good practicality in cross-scene HSI classification tasks, providing superior accuracy with acceptable training efficiency.

J. Inference Time Analysis

To evaluate the inference efficiency of different methods, Table X lists the test time on the Houston, S–H, and YC–YC datasets. This comparison reflects the practical computational cost during deployment.

As shown in Table X, TPMNet maintains competitive test times across all datasets. On the Houston and S–H datasets, its inference cost is significantly lower than DSAN, TSTnet, and TASA-Net while being comparable to lightweight methods such as FDDAN and MLUDA. On the YC–YC dataset, TPMNet also maintains a relatively low test time among the compared methods. These results demonstrate that the proposed framework does not introduce excessive computational burden during inference and remains suitable for practical large-scale HSI classification scenarios.

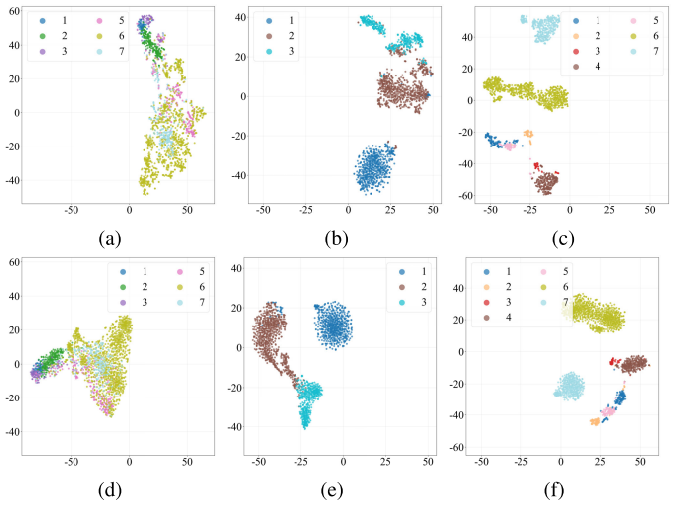


Fig. 10. t-SNE visualization of target-domain feature distributions before and after TPMNet training. (a) Houston dataset before training. (b) S–H dataset before training. (c) YC–YC dataset before training. (d) Houston dataset after training. (e) S–H dataset after training. (f) YC–YC dataset after training. Different colors represent different land-cover classes.

K. Feature Distribution Visualization

To intuitively demonstrate the effectiveness of TPMNet in learning discriminative cross-domain representations, we employ t-SNE figures to visualize the feature distributions of target-domain samples before and after training. Fig. 10 illustrates the t-SNE visualizations for three datasets, where the first row shows feature distributions before training and the second row presents distributions after TPMNet training.

As observed from Fig. 10, after TPMNet training, the target-domain feature distributions of all three datasets demonstrate significant improvements in class separability and intraclass compactness. For the Houston dataset, comparing Fig. 10(a) and (d), classes 6 and 7 exhibit notably more concentrated feature distributions after training, with clearer boundaries separating them from other classes. For the S–H dataset, Fig. 10(b) and (e) reveals that class 3 achieves substantially tighter clustering after adaptation, with samples

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TABLE XI
COMPARISON OF THE ROBUSTNESS OF LLM-GENERATED TEXT PRIORS

LLM	Houston datasets	S-H datasets	YC-YC datasets
Qwen-Plus	77.01 ± 1.37	**93.17 ± 0.07**	93.13 ± 0.77
DeepSeek-V3	75.95 ± 0.45	91.66 ± 0.14	**93.36 ± 0.38**
Kimi-K2	**77.09 ± 0.88**	91.65 ± 0.14	93.35 ± 0.20

forming a more cohesive cluster in the feature space. For the YC–YC dataset, the comparison between Fig. 10(c) and (f) shows that classes 6 and 7 demonstrate markedly improved intraclass concentration, with scattered samples being effectively grouped. These visualization results effectively bridge the domain gap through its dual-agent prior guidance and bidirectional multimodal alignment mechanisms.

L. Comparison of the Robustness of Different LLM-Generated Text Priors

To further verify the robustness of the proposed HSI-to-text generation, this section replaces the original Qwen-Plus model with DeepSeek-V3 and Kimi-K2 while keeping other parameters unchanged. The comparative experimental analysis is shown in Table XI.

As shown in Table XI, although the text descriptions generated by different LLMs are not identical, TPMNet still achieves consistently strong performance and outperforms the compared methods across three datasets. The slight performance variations imply that the proposed framework is not overly sensitive to the exact wording of the text priors. This is mainly because the two weighting strategies operate jointly during optimization. The similarity weight assigns higher importance to the text prior that is better aligned with the current visual feature. The historical weight keeps the semantic information stable across iterations and reduces the influence of less reliable descriptions from a specific LLM. This demonstrates TPMNet's robustness to text quality and its low sensitivity to the choice of LLM.

IV. CONCLUSION

To address the issues of insufficient prior semantic information and inadequate learning between image and text modalities in cross-domain HSI classification, this article proposes a dual-agent prior-guided multimodal network. In particular, to tackle the lack of prior information, a dual-agent collaborative cross-domain semantic modeling mechanism is designed, enabling cross-domain-aware dynamic prior text representations, so that textual information can play a more effective role in cross-domain HSI classification tasks. Results on the Houston target domain dataset show that TPMNet achieves more accurate classification for similar categories and complex, elongated distributions. On the YC–YC target domain dataset, TPMNet demonstrates superior classification accuracy across multiple categories, performing well on both regularly distributed and fragmented samples. These results indicate that TPMNet's dynamic prior texts enhance the effectiveness of cross-domain image classification. Meanwhile, existing multimodal methods often lack sufficient alignment between modalities, leading to underutilization of complementary information. The proposed BMAE introduces bidirectional cross-modal attention interactions, effectively capturing discriminative information from both image and text modalities, while a gating mechanism suppresses redundant features and improves the discriminative power of multimodal cross-domain features. Experimental results demonstrate that TPMNet outperforms existing methods on three cross-domain hyperspectral benchmark datasets. Therefore, the proposed approach effectively addresses the issues of insufficient prior information utilization and inadequate intermodal learning. Future work will explore the potential of this framework in larger-scale cross-domain tasks and broader multimodal remote sensing understanding applications.

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